

Skill Week - 7

1. To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs

```
GNU nano 8.7.1 src/input_redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {
    char input[100];
    int fd, saved_stdin;

    printf("Enter input file name: ");
    scanf("%99s", input);

    fd = open(input, O_RDONLY);

    if (fd < 0) {
        perror("Error opening input file");
        return 1;
    }

    saved_stdin = dup(STDIN_FILENO);

    dup2(fd, STDIN_FILENO);
    close(fd);

    printf("Reading redirected input:\n");

    char buffer[100];
    ssize_t n = read(STDIN_FILENO, buffer, sizeof(buffer) - 1);

    if (n > 0) {
        buffer[n] = '\0';
        printf("%s", buffer);
    }

    dup2(saved_stdin, STDIN_FILENO);
    close(saved_stdin);

    printf("\nInput stream restored.\n");

    return 0;
}
```

Output:

```
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ gcc -o input_redirect input_redirect.c
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ ./input_redirect
Enter a new file name: input.txt
Reading from input file
Hello from input file!

akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ ./input_redirect
Enter a new file name: abc.txt
Error opening input file: No such file or directory
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ |
```

2. To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection

```
GNU nano 8.7.1 src/output_redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    char filename[100];
    int fd, saved_stdout;

    printf("Enter output file name: ");
    scanf("%99s", filename);

    fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {
        perror("Error opening output file");
        return 1;
    }

    saved_stdout = dup(STDOUT_FILENO);

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("This text is redirected to the output file.\n");
    printf("Output redirection successful.\n");

    fflush(stdout);

    dup2(saved_stdout, STDOUT_FILENO);
    close(saved_stdout);

    printf("Output stream restored.\n");

    return 0;
}
```

Output:

```
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ nano src/output_redirect.c
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ gcc output_redirect.c -o output_redirect
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ ./output_redirect
Enter a filename to redirect output: output.txt
Output redirected successfully.

akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ cat output.txt
This text is redirected to the output file.
Output redirection successful.

akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ ls -l output.txt
-rw-r--r-- 1 akhil akhil 52 Apr 11 20:27 output.txt
```

3. To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

```
GNU nano 8.7.1 src/append_redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {
    char filename[100];
    int fd, saved_stdout;
    char *text = "New line appended to the file.\n";

    printf("Enter output file name: ");
    scanf("%99s", filename);

    fd = open(filename, O_WRONLY | O_CREAT | O_APPEND, 0644);

    if (fd < 0) {
        perror("Error opening file");
        return 1;
    }

    saved_stdout = dup(STDOUT_FILENO);

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("%s", text);
    fflush(stdout);

    dup2(saved_stdout, STDOUT_FILENO);
    close(saved_stdout);

    printf("Data appended successfully.\n");

    return 0;
}
```

Output:

```
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ cd ~/OS_Skill/Skill-Week-7
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ gcc src/append_redirect.c
-o bin/append_redirect
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ echo "Original data" > ap
pend.txt
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ cat append.txt
Original data
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ ./bin/append_redirect
Enter output file name: append.txt
Data appended successfully.
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ cat append.txt
Original data
New line appended to the file.
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ ./bin/append_redirect
Enter output file name: append.txt
Data appended successfully.
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ cat append.txt
Original data
New line appended to the file.
New line appended to the file.
```

4. To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

```
GNU nano 8.7.1 src/stderr_redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd, saved_stderr;

    fd = open("error.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {
        perror("Cannot open error.txt");
        return 1;
    }

    saved_stderr = dup(STDERR_FILENO);

    dup2(fd, STDERR_FILENO);
    close(fd);

    FILE *file = fopen("missing.txt", "r");

    if (file == NULL) {
        perror("Error opening missing.txt");
    } else {
        fclose(file);
    }

    fflush(stderr);

    dup2(saved_stderr, STDERR_FILENO);
    close(saved_stderr);

    printf("stderr restored.\n");

    return 0;
}
```

Output:

```
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ gcc src/input_redirect.c
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ ./bin/input_redirect
Enter input file name: input.txt
Reading redirected input:
Hello from input file!

akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ gcc src/output_redirect.c
akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ ./bin/output_redirect
Enter output file name: output.txt
Output stream redirected.

akhil@Akhil-laptop:~/OS_Skill/Skill-Week-7$ cat output.txt
This text is redirected to the output file.
Output redirection successful.
```